

Hypothesis Testing I

Rex Shao-Yu Wang

Department of Agronomy, National Taiwan University

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Outline

Hypothesis Testing

One sample z-test

One sample t-test

Two sample z-test

Lab Report

Type I and type II errors

Table of error types		Null hypothesis (H_0) is	
		True	False
Decision about null hypothesis (H_0)	Not reject	Correct inference (true negative) (probability = $1 - \alpha$)	Type II error (false negative) (probability = β)
	Reject	Type I error (false positive) (probability = α)	Correct inference (true positive) (probability = $1 - \beta$)

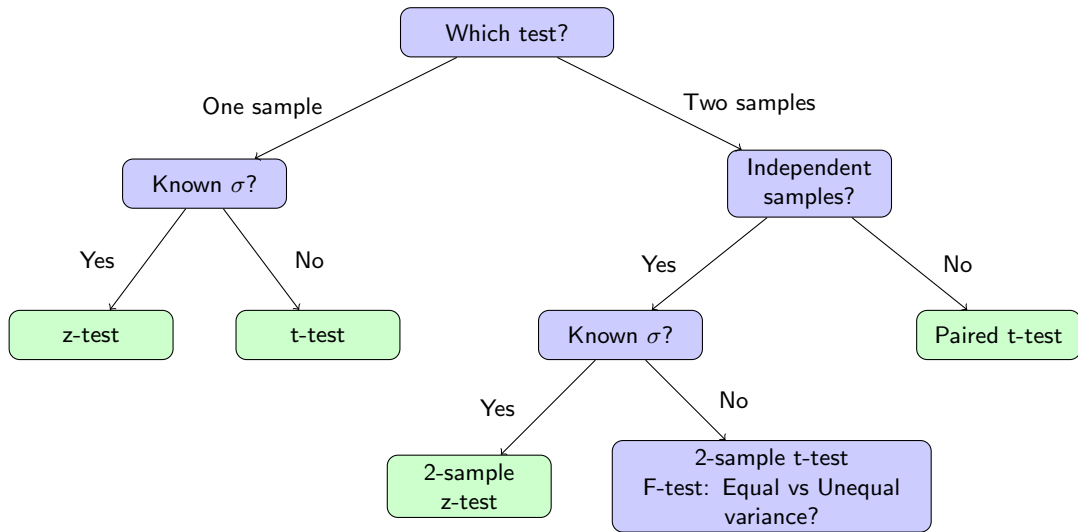
Procedure for Hypothesis Testing

1. State your null and alternate hypothesis
2. Define significant level α
3. Decide the test statistic (Z, T, F)
4. Find the rejection region (RR)
5. Calculate your test statistic and compare with RR
6. Make a decision whether reject H_0 or not

Load the package BSDA before we start.

```
library(BSDA)
```

Types of Hypothesis Testing



One-tailed vs Two-tailed

One sample z-test

- ▶ Hypothesis:

- ▶ Two-sided: $H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$
- ▶ One-sided upper-tail: $H_0 : \mu \leq \mu_0$ vs. $H_a : \mu > \mu_0$
- ▶ One-sided lower-tail: $H_0 : \mu \geq \mu_0$ vs. $H_a : \mu < \mu_0$

- ▶ Test Statistic:

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \stackrel{H_0}{\sim} N(0, 1)$$

- ▶ Rejection Region (RR):

$$RR = \{|Z| \geq z_{\frac{\alpha}{2}}\} = \{Z \geq z_{\frac{\alpha}{2}} \text{ or } Z \leq -z_{\frac{\alpha}{2}}\}$$

One sample z-test

Let significance level $\alpha = 0.05$, sample mean = 24, known population variance = 100

- ▶ Two-sided: $H_0 = \mu_0$ vs. $H_a \neq \mu_0$

```
z_test <- z.test(data, mu = 24, sigma.x=10, conf.level =  
0.95, alternative = "two.sided")
```

- ▶ One-sided upper-tail: $H_0 \leq \mu_0$ vs. $H_a > \mu_0$

```
z_test <- z.test(data, mu = 24, sigma.x=10, conf.level =  
0.95, alternative = "greater")
```

- ▶ One-sided lower-tail: $H_0 \geq \mu_0$ vs. $H_a < \mu_0$

```
z_test <- z.test(data, mu = 24, sigma.x=10, conf.level =  
0.95, alternative = "less")
```

One sample t-test

- ▶ Hypothesis:

- ▶ Two-sided: $H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$
- ▶ One-sided upper-tail: $H_0 : \mu \leq \mu_0$ vs. $H_a : \mu > \mu_0$
- ▶ One-sided lower-tail: $H_0 : \mu \geq \mu_0$ vs. $H_a : \mu < \mu_0$

- ▶ Test Statistic:

$$T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}} \stackrel{H_0}{\sim} t(n-1)$$

- ▶ Rejection Region (RR):

$$RR = \{|T| \geq t_{\frac{\alpha}{2}}\}(n-1) = \{T \geq t_{\frac{\alpha}{2}}(n-1) \text{ or } T \leq -t_{\frac{\alpha}{2}}(n-1)\}$$

One sample t-test

Let significance level $\alpha = 0.05$, sample mean = 24, unknown population variance

- ▶ Two-sided: $H_0 = \mu_0$ vs. $H_a \neq \mu_0$

```
t_test <- t.test(data, mu = 24, conf.level = 0.95,  
                 alternative = "two.sided")
```

- ▶ One-sided upper-tail: $H_0 \leq \mu_0$ vs. $H_a > \mu_0$

```
t_test <- t.test(data, mu = 24, conf.level = 0.95,  
                 alternative = "greater")
```

- ▶ One-sided lower-tail: $H_0 \geq \mu_0$ vs. $H_a < \mu_0$

```
t_test <- t.test(data, mu = 24, conf.level = 0.95,  
                 alternative = "less")
```

Two sample z-test

- ▶ Hypothesis:

$$\begin{cases} H_0 : \mu_1 = \mu_2 \\ H_A : \mu_1 \neq \mu_2 \end{cases}$$

- ▶ Test Statistic:

$$Z = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \stackrel{H_0}{\sim} N(0, 1)$$

- ▶ Rejection Region:

$$RR = \{|Z| \geq z_{\frac{\alpha}{2}}\} = \{Z \geq z_{\frac{\alpha}{2}} \text{ or } Z \leq -z_{\frac{\alpha}{2}}\}$$

Two sample z-test

We get two data, data1 and data2, knowing that $\sigma_1 = 10, \sigma_2 = 15$.

We want to test whether $\mu_1 = \mu_2$ with significance level $\alpha = 0.05$?

```
data1 <- c(27, 24, 18, 29, 30, 27)
```

```
data2 <- c(23, 28, 20, 19, 35, 23)
```

```
z_test <- z.test(data1, data2, mu = 0, sigma.x = 10, sigma.y  
  = 15, conf.level = 0.95)
```

Lab Report

- ▶ The report file will be in the NTU COOL Assignment section
- ▶ Submit your file in pdf format
- ▶ Deadline: 5/3 (Sun) 23:59
- ▶ Late submission is not accepted